

Chapter 160

Resolve FX Light

The plugins in this category all replicate different optical and lighting effects. While most have been designed to quickly give you realistic results, all can be pushed harder to provide many artistic effects.

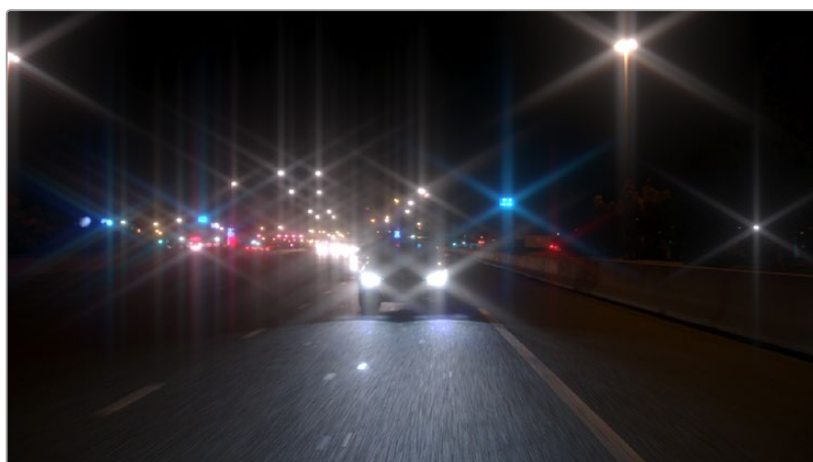
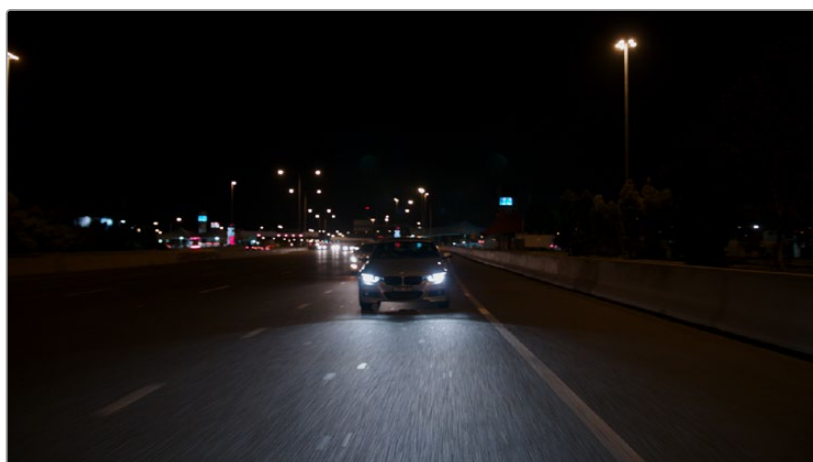
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Aperture Diffraction (Studio Version Only)

Found in the ResolveFX Light category, Aperture Diffraction models the starburst effect usually seen when shooting bright lights with small apertures, the physical cause of which is light-diffraction on the aperture blades of a lens. This plugin simulates this, with the result being automatically applied to scene highlights that you can isolate and refine, with customizable virtual apertures.

Small regions of brightness exhibit a star pattern glow, as seen in the following image.



(Top) Original image, (Bottom) Applying Aperture Diffraction

Large regions of brightness exhibit a more even glow with shaping and texture that look like a natural optical effect. It can be used to create a different type of glow effect with a more realistic look in some situations than the Glow plugin, though it's more processor intensive. In other situations, this plugin opens up many different stylistic possibilities for glowing effects.



(Top) Original image, (Bottom) Applying Aperture Diffraction

Output

These controls let you choose what image is output by this plugin.

- **Select Output:** Lets you preview the image with different stages of the Aperture Diffraction effect applied.
 - Isolated Source:** Used to help when adjusting the Isolation controls.
 - Preview Aperture:** Used to help when adjusting the Aperture controls.
 - Preview Diffraction Pattern:** Shows you the resulting diffraction pattern based on the aperture control settings.
 - Diffraction Patterns Alone:** Shows you the glow effect that will be applied to the image by itself.
 - Final Composite:** Shows you the final output of the effect.
- **Alpha Masks Light Sources:** Check this box to use the input Alpha channel to exclude sources in those areas from casting light rays.
- **Alpha Limits Effect:** Check this box to use the input Alpha channel for compositing purposes. Unchecking means the input Alpha is ignored.
- **Quality:** Lets you choose a quality setting to trade off between quality and speed. Choices are Full, Half (Faster), Quarter (Fast). High quality gives diffraction of finer detail, but otherwise does not change the effect. If you have no small light point sources, you can safely use half or quarter qualities with no reduction in image quality.

Isolation Controls

The Isolation controls let you choose which highlights in the scene generate visible glow and patterns. The effect of these controls can be directly monitored by setting Select Output to Isolated Source.

- **Color Mode:** A pop-up menu that lets you either choose to keep the colors of the different highlight regions that generate glow, or treat them all as grayscale brightness only (color controls later can change the effect). Grayscale is faster to process, but Color can result in some brilliant effects.
- **Brightness:** Sets the threshold at which highlights are isolated.
- **Gamma:** Lets you shape the isolated highlights.
- **Smooth:** Lets you blur details in the highlights that you don't want to be pronounced.
- **Color Filter:** Lets you choose a particular color of highlight to isolate (an eyedropper lets you select a value from the Viewer).
- **Operation:** Lets you adjust the resulting Isolation matte (options include Shrink, Grow, Opening, Closing) with a slider to define how much.

Aperture Controls

The Aperture controls let you define the shape and texture of the resulting glow this plugin creates.

- **Iris Shape:** Lets you choose a shape that determines how many arms the star pattern will have. Options are Triangle, Square, Pentagon, Hexagon, Heptagon, and Octagon.
- **Aperture Size:** Lets you alter the resulting diffraction pattern alternating between more of a star shape at higher values and a stippled wave pattern at lower values.
- **Blade Curvature and Rotation:** Lets you alter the softness and orientation of the arms of each star.
- **H/V Ratio:** Alters the horizontal/vertical ratio of the aperture, allowing you to replicate an anamorphic glow.
- **Angle:** Sets the angle of the aperture.
- **Chroma Shift:** Lets you introduce some RGB “bleed” into the glow.

Diffraction Controls

- **Result Gamma:** Lets you adjust how pronounced will be the glow that appears between the arms of the star patterns that appear.
- **Result Scale:** Lets you alternate between pronounced star patterns at high values and more diffuse glows at low values.

Compositing Controls

These controls let you adjust how to composite the glow effect against the original image.

- **Normalize Brightness:** This checkbox scales the brightness of the glow per frame. The Aperture Diffraction effect will keep to a consistent overall brightness as the scene changes.
- **Brightness:** Lets you adjust the intensity of the glow effect.
- **Colorize:** Lets you tint the glow effect using the Color control that appears below.

Glow

A sophisticated, soft glow effect that's highly customizable with only a few key parameters.

- **Select Output:** Lets you preview the image with different stages of the glow effect applied, viewing the Shiny Regions, the Glow Alone, or the Glowing Image.
- **Shine Threshold:** Defines the luminance level at which glow is triggered to appear on the image.
- **Alpha Masks Light Sources:** Check this box to use the input Alpha channel to exclude sources in those areas from casting light rays.
- **Alpha Limits Effect:** Check this box to use the input Alpha channel for compositing purposes. Unchecking means the input Alpha is ignored.

Shape and Spread

- **Spread:** Defines how far out shine extends from areas of the picture that trigger it.
- **H/V Ratio:** Lets you adjust the proportion of horizontal to vertical spread, allowing you to create "streaky" shine that extends farther in a particular dimension.
- **Relative Spread Red/Green/Blue:** Lets you adjust the spread within each color channel by a different amount, simulating a kind of chromatic aberration in the glow.

Color and Composite

- **Gain:** Adjusts the brightness of the glow effect.
- **Gamma:** Adjusts the spread of the glow effect and how it fades with distance.
- **Saturation:** Adjusts the intensity of the glow's color.
- **Color Filter:** A color picker and eyedropper let you choose a color with which to tint the glow.
- **Glow framing:** Lets you choose what happens when shine hits the edge of the frame, whether it's amplified by "Reflect in Camera" or whether it's moderated by a "Vignette" effect.
- **Composite Type:** Lets you choose a composite mode to use to blend the glow effect with the image. Defaults to Screen, which is good for gentle glows. Add lets you create hotter, more intense glows, while other composite modes let you create other varied effects.
- **Opacity:** Lets you adjust the transparency of the glow effect and is a fast way of "easing off" of a glow that you otherwise like, but that's too intense for the shot at hand.
- **Invert Effect:** Checking this box inverts the Glow parameters, so darker areas now affect brighter regions instead of the other way around.
- **Bright Region Recovery:** Adjust this slider to reclaim some image detail in areas that are blown out by the glow.

Lens Flare (Studio Version Only)

Simulates different kinds of lens flares resulting from the interplay of light through an aperture of selectable shape bouncing within layered optical elements in a lens. Four groups of controls give you a lot of ways to customize the presets to create your own effect:

Lens Flare Preset

You can choose a type of flare from the Lens Flare Preset drop-down. If you manipulate the controls to create your own effect, this drop-down switches to Custom.

Select Output

The Select Output drop-down lets you switch among the following viewing options:

- **Final Image:** (Default) Shows the lens flare composited against the current clip.
- **Flare Elements Alone:** Shows the isolated flare, making it easier to adjust.
- **Source Mask and Magnified Source Mask:** Let you see the mask used to control the lens flare brightness.

Light Source Masking

Three additional controls at the top let you create a quick and dirty luma key to create a mask that you can use for occluding the lens flare effect behind foreground elements in the scene. For example, if you're using lens flare to simulate the sun in the sky moving behind trees, these controls let you use those tree silhouettes to shift and modify the flare source accordingly.

- **Enable Light Source Masking:** This checkbox enables the Mask Threshold and Virtual Light Source Size sliders. You can then choose either Source Mask or Magnified Source Mask to see the mask that's created by raising the Mask Threshold slider.
- **Mask Threshold:** Raising this slider pulls a fast Luma Key that masks off darker foreground elements (such as trees against the horizon) and limits the flare to areas of the picture where the light source of the flaring can "shine through" (In this example, shining between the trees on the horizon).
- **Virtual Light Source Size:** This slider determines how quickly the center of a flare will disappear behind the luma-keyed mask you've created; higher values (simulating bigger light sources) cause a flare to disappear more slowly when passing "behind" a feature of the image that's being keyed via the Mask Threshold slider.

Position

Position parameters let you adjust the X and Y Position of the simulated light source that's causing the flare. There are three ways you can adjust the flare position.

- You can Adjust the X and Y Position sliders.
- You can turn on the OpenFX Overlay in the Viewer and drag the on-screen control.
- You can motion track elements within the scene using the FX Tracker.

Checking the Move With Sizing checkbox lets the lens flare retain its relative position in the frame if the Input and Editing sizing are changed.

Global Corrections

The Global Corrections group have parameters that let you quickly adjust the overall quality of the flare effect.

- **Global Scaling:** Lets you make every element of the flare effect larger or smaller at once.
- **Anamorphism:** Lets you stretch all flare elements horizontally to simulate an anamorphic lens' stretching effect.
- **Lens Center X and Y position:** Lets you offset the center of the simulated lens that's causing the flaring, about which the various elements of the flare pivot.
- **Global Defocus:** Lets you blur the overall flaring effect being created to soften the effect.
- **Global Brightness:** Lets you raise and lower the overall level of flaring being produced.
- **Global Saturation:** Lets you adjust the overall color intensity of the flaring.
- **Colorise Result:** Lets you choose how much to tint the flare, if at all. At a value of 0 (the default), no colorization is applied at all.
- **Colorization Color:** Lets you choose a color to use to colorize the flare via the Colorize Result slider above. You can use either a color control or eyedropper to sample color from the source RGB image of the current node.

Aperture

The Aperture parameters let you define the aperture of the simulated camera apparatus through which the flare is being generated. The shape defined by these parameters affects the look of the “starburst element” of each flare, as well as the look of any aperture-shaped “ghost elements” you're turning on.

- **Aperture Blades:** Defines how many blades make up the aperture. You can choose from 3 to 16 (defaults to 6). The diffraction of the lens flares is symmetric, so an odd number of blades will give you twice as many rays as the number of blades, while an even number of blades will give you the same amount of rays.
- **Angle:** Sets the angle of the resulting Aperture shape.

Elements

The Elements drop-down is a deceptively simple set of controls that let you expose the customization controls of each of the layers and elements that combine to create the simulated lens flare. Up to ten levels of elements can contribute to a lens flare effect. Each element and ghost shape has a different set of parameters unique to that type of element. Available elements in this pop-up menu include:

- **Full-Screen Glare:** A simulated overall flaring that covers the entire frame at its most intense. This glare increases as the flare nears the center of the screen and decreases as the flare moves toward the edges of the frame. Recommended for extremely big flaring effects. Parameters include:
Glare Brightness: A slider; set this to 0 to eliminate glare.
Glare Color: A color picker and eyedropper combination that lets you tint the glare.
- **Flare Spot:** Simulates the central light source that's triggering the flare. Parameters include Flare Size (set this to 0 to eliminate the flare spot), Flare Irregularity (which lets you create a more organic-looking, off-balance flare), Flare Softness, and Flare Color.
- **Starburst:** Rays of light stretching out from the center of the flare. Parameters include Starburst Size (set this to 0 to eliminate the starburst), Starburst Softness, Starburst Split Angle (which splits each streak of the starburst into a wider feathered pattern), Starburst Split Balance (which lets you adjust the brightness between each half of split streaks), and Starburst Color.

- **Ghost elements:** Seven available layers of Ghost Elements can be enabled and set to different optional shapes for each lens element you want to simulate. Each of the Ghost element layers can use one of five types of shapes, including:

None: Turns that particular ghost element off.

Aperture Shape: A polygonal shape defined by the number of blades in the aperture you've selected.

Anamorphic Streak: A wide horizontal artifact typical of anamorphic lenses.

Disc Shape: A round ring artifact.

Bubble Shape: An oval with haze within.

Corona Rays: A ring of streaks stretching outward.

Ghost elements share many parameters in common, although specific elements have unique parameters. These parameters are as follows:

Color: A color picker and eyedropper combination that lets you colorize that specific element.

Position: A slider lets you set that element's position along the optical path that's defined by the angle from the (Flare) Position X and Y to the Lens Center X and Y parameters. A value of 0 centers that element on the Lens Position, while larger values push that element farther and farther away from the Lens Position.

Size: Sets the size of that flaring element, or in the case of the Anamorphic Streak, the width.

Height: (Anamorphic Streak only) Sets the vertical thickness of the streak.

Center Brightness: (Aperture, Disk, and Anamorphic Streak only) Defines the brightness in the middle of that element, filling it to appear as a solid element. Set this closer to 0 if you want the element to appear hollow.

Edge Brightness: (All but Anamorphic Streak) Defines the brightness of the edge of that element. Raising this value while lowering the Center Brightness lets you create outlined element effects.

Softness: Lets you blur that element.

Bristle Density: (Corona Only) Lets you alter the number and arrangement of the optical bristles that appear. At lower values, fewer bristles appear, at higher values, more bristles appear. As you change the value of this parameter, the placement of bristles shifts, allowing you to change distribution as well as density with a single control.

Bristle Scale: (Corona only) Lets you alter the thickness of the bristles that appear. Smaller values result in thicker bristles, higher values result in smaller bristles more tightly packed together.

Ringings: (All types but Bubble) Simulates a pattern of diffraction artifacts. Higher values increase the number of rings or streaks comprising the element.

Chromatic shift: Simulates chromatic aberration effects.

Eclipse Position: Simulates where the outer (away from the center) or inner (toward the center) side of a Ghost Element is missing because the light is off the edge of a lens element or blocked by some part of the tube housing. Practically, this defines which side of the Ghost Element is affected by the other Eclipse parameters of size, softness, and chromatic shift. Adjusting this results in semi-circular ghost shapes of different kinds. At 0, there is no eclipse. At positive values, the eclipse starts from within the frame and pushes out; at negative values, the eclipse starts from the outside of the frame and pushes in.

Eclipse Size: Defines the size of the eclipse region for that flare element. At higher values, more and more of the flare element is eclipsed.

Eclipse Softness: Defines the softness at the transition from the eclipsed and non-eclipsed regions.

Eclipse Chromatic shift: Lets you create a chromatic aberration effect at the boundary of the eclipsed region. At 0, there is no chromatic shift. At increasingly positive values, there's a shift toward blue. Toward negative values, there's a shift toward red.

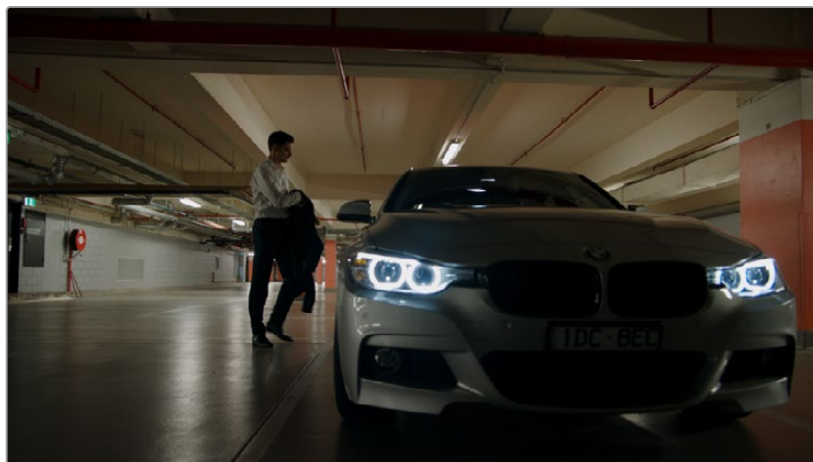
Repeat: Lets you use this element to spawn many duplicates defined by the following two parameters.

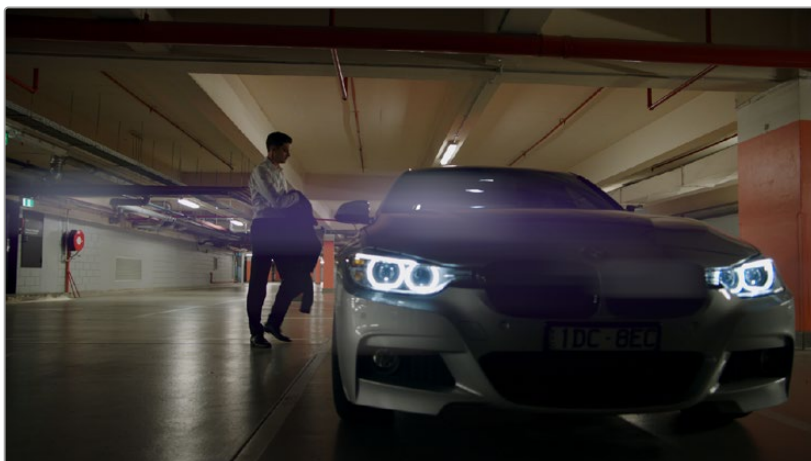
Repeat Position Seed and Repeat Size Seed: At different values, these parameters let you pseudo-randomly redistribute the placement of repeated elements.

Lens Reflections (Studio Version Only)

Found in the ResolveFX Light category, Lens Reflections simulates intense highlights reflecting off the various optical elements within a lens to create flaring and scattering effects based on the shape and motion of highlights you isolate in the scene. It's an effective simulation that works best when there are light sources or specular reflections in the scene such as the sun, car headlights, light fixtures, fire and flame, or other lighting elements that are plausibly bright enough to cause such flaring.

Also, this plugin really shines when these light sources move, as each layer of simulated reflections moves according to that element's position within the virtual lens being simulated, creating organic motion that you don't have to keyframe. Without intense highlights, the results of this filter are good for more abstract light-leak effects.





(Previous page) Original image, (Above) Applying Lens Reflections

Output

The Output controls let you preview the image with different stages of the Lens Reflections effect applied.

- **Select Output:** Lets you choose to view the Isolated Source (to help when adjusting the Isolation controls), Reflections Alone (showing you the flaring effect that will be applied to the image by itself), or the Final Composite (the complete effect).
- **Alpha Masks Light Sources:** Check this box to use the input Alpha channel to exclude sources in those areas from casting light rays.
- **Alpha Limits Effect:** Check this box to use the input Alpha channel for compositing purposes. Unchecking means the input Alpha is ignored.
- **Quality:** This pop-up lets you choose how to render the effect. Options are Full, Half (Faster), and Quarter (Fast). The tradeoff is between sharpness and speed. The actual reflection process is the same across all quality levels. Smoother and lower resolution content will not benefit from the Full Quality setting.

Isolation Controls

The Isolation controls let you choose which highlights in the scene will generate lens reflections. The effect of these controls can be directly monitored by setting Select Output to Isolated Source. It's highly recommended to customize the Isolation controls for the image at hand when using this plugin, as even more so than other plugins, the particular highlights used will have a huge impact on the resulting effect.

- **Color Mode:** A pop-up menu that lets you either choose to keep the colors of the different highlight regions that generate lens reflections, or treat them all as grayscale brightness only (color controls later can change the effect). Grayscale is faster to process, but Color can result in some brilliant effects.
- **Brightness:** Sets the threshold at which highlights are isolated.
- **Gamma:** Lets you shape the isolated highlights.
- **Smooth:** Lets you blur details in the highlights that you don't want to be pronounced.
- **Color Filter:** Lets you choose a particular color of highlight to isolate (an eyedropper lets you select a value from the Viewer).
- **Morph Operation:** A pop-up lets you adjust the resulting Isolation matte (options include Shrink, Grow, Opening, Closing), and a slider lets you define how much.

Morph Amount: Adjusts the edges of the light source based on the Morph Operation chosen.

Global Controls

The Global controls let you quickly and easily adjust the overall quality of the Lens Reflections effect using a centralized group of parameters.

- **Global Brightness:** Lets you raise and lower the level of all reflections. Lowering brightness is a good way to make a large lens reflection effect more subtle, although images with small lens reflection effects may benefit from being a little brighter.
- **Global Blur:** Lets you defocus all reflections. This is another good way of making lens reflection effects of all kinds more subtle.
- **Anamorphism:** Lets you deform the reflection elements to simulate an anamorphic lens' stretching effect.
- **Global Colorize:** Lets you adjust the color intensity of the reflections, either intensifying the color of all reflections or desaturating it.

Presets

A Presets pop-up provides a number of different settings to get you started. Selecting a preset populates the Reflecting Elements parameters below, at which point you can customize the effect to work best with the image at hand. It's highly recommended to customize these effects to suit the type of highlights in your image, in order to get the best results.

Reflecting Elements

There are four groups of Reflecting elements, each with identical controls. This lets you create interactions combining up to four sets of reflections. The controls found within each group are as follows.

- **Brightness:** Lets you adjust the intensity of that reflection.
- **Position in Optical Path:** Lets you shift the reflection according to an element's position in the lens. Practically, this means that positive values will enlarge an inverted reflection based on the highlights, while reducing values toward 0 will shrink the reflection, and pushing this into negative values will invert the reflection and pull it into the opposite direction as it begins to enlarge again. A value of -1 positions the reflection right over the highlight that creates it.
- **Defocus type:** Lets you choose what kind of blur to use, choices include Box blur, Triangular blur, Lens blur (the most processor intensive), and Gaussian blur (the default).
- **Defocus:** Lets you choose how much to blur that element.
- **Stretch:** Lets you give the flare an anamorphic widescreen look.
- **Stretch Falloff:** Lets you taper the edges for a less uniform stretching effect.
- **Lens Coating:** A pop-up lets you choose common colors such as purple, green, and yellow that correspond to different anti-reflective lens coatings, as well as a selection of other vibrant colors. Defaults to none. When you choose any other option, a color control and eyedropper let you manually choose a color or pick one from the image. A Colorize slider lets you vary how much to tint the reflection by the selected color, although setting Colorize to 0 lets the flare take its color from the source highlights of the image, which can sometimes give you the most interesting look.

Light Rays

A “rays of light” effect that simulates volumetric lighting emerging from light sources defined by a threshold you define. The effect mimics what are sometimes called “god rays” in the sky, or other highly directional glow effects.

Main Controls

The Main controls include:

- **Select Output:** Lets you preview the image with different stages of the Lightray effect applied, viewing the Final Image, Lightrays Alone, and Source Regions.
- **Source of Rays:** Lets you choose the emitter of the rays, using Bright Regions or Edges.
- **Source Threshold:** A slider that lets you choose the limit at which bright areas of the image emit rays.
- **Alpha Masks Light Sources:** Check this box to use the input Alpha channel to exclude sources in those areas from casting light rays.
- **Alpha Limits Effect:** Check this box to use the input Alpha channel for compositing purposes. Unchecking means the input Alpha is ignored.

Position

Position parameters let you define the direction rays take.

- **Ray Directions:** A pop-up menu that lets you choose either “From A Location,” which exposes controls to let you choose an X or Y position to define a point of origin that defines the angle of the light beams, or “At an Angle,” which exposes controls to let you choose an overall orientation for the rays.

Appearance

Appearance parameters let you customize the ray effect.

- **Ray Dropoff:** A pop-up with four options.
 - Default (soft):** Produces soft, indistinct rays of light that appear to fade away as they stream out.
 - Keep Shape of Source:** The edge of the light rays are defined by the shape that emitted them.
 - CCD Bloom Harsh:** Severely raises the brightness of the part of the image that’s emitting rays as Length is raised, resulting in harsh glow or bloom in the image.
 - CCD Bloom Soft:** Gently raises the brightness of the part of the image that’s emitting rays as Length is raised, resulting in a very gentle lightning of the image.
- **Length:** Lets you make the rays longer or shorter.
- **Soften:** Lets you blur the rays being emitted.
- **Brightness:** Lets you adjust how bright the rays are.
- **Gamma:** Adjusts the brightness of the ray effect using a gamma curve.
- **Saturation:** Adjusts the intensity of the ray’s color.
- **Color:** A color picker and eyedropper let you define a color with which to tint the rays.
- **Composite Type:** Lets you choose a composite mode to use to blend the glow effect with the image. Defaults to Add, which is good for hotter, more intense rays. Screen lets you create more gentle rays, while other composite modes let you create other varied effects.
- **Bright Region Recovery:** Adjust this slider to reclaim some image detail in areas that are blown out by the rays.